



COLLEGE OF ENGINEERING AND INFORMATION TECHNOLOGY
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Case Study #2 – Air Cargo's e-Enterprise System

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Air Cargo, Inc. (ACI) is a logistics management company providing air cargo ground services to 17+ shareholder airlines and 53 associates, including a road feeder service that connects airports and cities in the United States and Europe, pickup and delivery of regular air cargo shipments, small package air cargo pickup, and delivery of flight specific, time-definite shipments. A major system crash in the late 1990s caused ACI's revenue to drop from about \$150 million to about \$145 million; this served as a wake-up alarm for management and started their systems integration effort with their e-Enterprise system.

PROBLEMS WITH FUNCTIONAL SILOS

ACI runs various business applications for dispatching, invoicing, freight audits, and reconciliation. Before converting to ERP, ACI's accounting staff had to export and import text files to communicate across accounting and business applications. This took time and made even the most recent report slightly dated. ACI initially considered simply upgrading its existing accounting system to a new release; however, Jack Downing, ACI's IT director, decided this approach was not feasible because they had developed numerous custom procedures in their accounting and other business applications. "An upgrade wouldn't have been economical," he explained, "because we had made so many custom changes to the old software."

e-ENTERPRISE SYSTEM

ACI therefore implemented four e-Enterprise applications—its financial series, distribution series, customization series, and integration series—and integrated them with a database management system, Microsoft SQL Server, at the backend. In addition, ACI used the Microsoft transformation services. The immediate effect of this implementation was the integration of the accounting system with the line-of-business applications, thereby eliminating manual data reentry. This occurred because Microsoft SQL Server and the Data Transformation Services (DTS) software connected to ACI's line-of-business applications and delivered transaction data to the ACI e-Enterprise system. DTS software is an extract, transform, and load (ETL) tool which consolidates data from disparate sources to single or multiple destinations. These data were used to update the financial records. Note that DTS runs in the background on a regular basis without requiring manual intervention. As a result, the accounting department no longer needs to export and import line-of-business records manually to generate its reports.

BENEFITS

The key advantage of the e-Enterprise system is that it links contracts managed in the line-of-business applications to financial records. "Once we agree on rates, terms, and hours of operation with a contractor, they are entered into our contracts administration system," says Sally Hartmann, ACI's CFO. "In the past, we had to reenter that information into the financial system manually and hope that we remembered to adjust it if there were changes in the contract. But now, we automatically move this information from the accounting system to the contract management system. And we have the ability to move that information in the other direction as well, which we didn't have in the past." The key benefit here is that if a contractor stops paying its bills, the warehouse system will immediately alert the dispatch people. Hartmann adds: "We are also completing a direct link from our human resources to our accounting system. This will eliminate another area of manual data entry and reduce the potential for errors." Hartman ultimately sees three other advantages accruing to ACI from this ERP conversion. These are as follows:

- "Based on what I have seen so far, I am estimating that we will reduce our administrative workload by about 10 percent, primarily by reducing manual data entry."
- "With the ERP system, we will provide more accurate, complete, and timely information to decision makers."
- "We are moving heavily into e-commerce, and e-Enterprise provides the building blocks that enable two-way information flow between e-business and financial systems."



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The ACI case highlights some of the problems with heterogeneous systems and how they can affect the various departments within the company from Accounting to Warehousing. The systems integration at ACI has introduced some benefits and drawbacks.

Questions:

1. Provide insights on System Integration on ACI's move to an integrated e-Enterprise System.

ACI is a big and successful logistics company that operates across continents. Before the late 1990s, it used different business systems that were able to get the job done. Unfortunately, after a major system crash caused the company to lose a significant amount of revenue, ACI realized that although its existing system had helped the company succeed, it also had several limitations. One of the biggest problems they faced was that the accounting staff had to manually export and import files just to share information with other company systems. This process took a lot of time, increased the chances of human error, and caused reports to become slightly outdated by the time they were generated. Instead of upgrading the old accounting system, ACI implemented an integrated e-Enterprise system because the previous software had too many custom changes. By integrating the system with Microsoft SQL Server and DTS, information could be shared automatically between different applications, so employees no longer had to enter the same data over and over again. This made their work faster and helped keep the information accurate and updated across different departments. The integration also made it easier for departments to work together since everyone could access the same information whenever changes were made. As a result, there was less manual work, fewer human errors, and managers could rely on up-to-date reports when making decisions. Although ACI only decided to improve its system after experiencing a major problem, the move to an integrated e-Enterprise system helped make its operations more efficient and better connected across different departments.

2. What are the advantages of moving from traditional system to Integrated System?

Compared to a traditional system, an integrated system makes it easier for a company to manage its operations and share information across different departments. In a traditional system, different departments use separate applications, so employees have to manually transfer or reenter data, which takes time, increases the risk of errors, and can result in outdated information. This was one of the problems experienced by ACI before implementing its e-Enterprise system. With an integrated system, information is stored in one centralized database and can be accessed by different departments whenever it is needed. In ACI's case, the e-Enterprise system automatically shared information between different systems, making records more consistent and removing the need for repeated data entry. Another advantage is that all departments can work with the same updated information, making it easier to coordinate tasks and complete processes without unnecessary delays. Managers also have access to more reliable reports, allowing them to monitor operations more effectively and respond more quickly when issues arise. In the long run, an integrated system helps companies work more smoothly by simplifying daily processes and allowing employees to focus on more productive tasks instead of repetitive administrative work.

3. Identify the problems with Functional Silos.

One of the main issues with ACI's functional silos was that different departments were using separate systems that were not connected to each other. Because of this, employees had to manually export and import data between applications, which took a lot of time and increased the chances of making mistakes. Another problem was that whenever changes were made in one system, such as updating contract information, employees also had to manually update the other system. This increased the risk of inconsistent data and missing important updates. Since information was not shared automatically, reports could easily become outdated, making it harder for managers to make timely decisions. It also slowed down communication between departments because information had to be transferred manually instead of being shared in real time.



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4. As a future Software Engineering and IT Expert, what are your solutions to ACI's functional silos? How would you plan out your solution to some of the problems you identified in question #3?

I would recommend using an integrated ERP system that allows all departments to share information through one centralized database. This would remove the need for manual data transfer and ensure that everyone has access to accurate and updated information. I would also automate the movement of data between different systems to reduce human errors and improve the speed of daily operations. To implement this solution, I would first study ACI's existing systems and identify which departments and systems need to be connected. After that, I would plan the integration carefully and test the system before fully deploying it to make sure everything works properly. I would also provide training for employees so they can easily adapt to the new system. Finally, I would continuously monitor the system, gather feedback from users, and make improvements whenever needed to ensure that it continues to support the company's daily operations and future business needs.